QPCODE-25901713

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCIAM100 - INTELLIGENT AUTOMATION TECHNIQUES

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 **Minutes Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A Multiple Choice Questions Answer all questions Each question carries 1 marks

1. Which of the following is not an application of artificial intelligence?
   d). Data Base Management System [U] / [CO1]
2. How do you install Jupyter Notebook?
   b). pip install notebook [U] / [CO2]
3. What happens if too many clusters are formed in hierarchical clustering?
   d). The model loses interpretability [U] / [CO3]
4. Which of the following is an application of Machine Learning in medical treatment?
   d). All of the above [U] / [CO1]
5. Which method is used for regression in KNN?
   c). Taking the average of K nearest neighbors [U] / [CO3]
6. What does unsupervised learning aim to do?
   b). Find hidden patterns or intrinsic structures in input data [U] / [CO1]
7. What happens when gradient descent reaches a saddle point?
   a). The gradient becomes zero, and progress slows down [U] / [CO2]
8. Which of the following is a key characteristic of unsupervised learning?
   d). It works without labeled data [U] / [CO3]
9. What does the regression line represent in linear regression?
   c). Best fit line for predicting output [U] / [CO3]
10. Which of the following is a subset of Artificial Intelligence?
    d). All of the above [U] / [CO1]

[1x10 = 10]

Part B

Short Answer Questions. Answer any 4 questions Each question carries 5 marks

11. Explain the relationship between AI, Machine Learning, and Deep Learning. [U] / [CO1]
    - *Artificial Intelligence (AI) is the broad field of creating intelligent systems.*
    - *Machine Learning (ML) is a subset of AI that enables systems to learn from data.*
    - *Deep Learning (DL) is a subset of ML that uses neural networks with many layers.*

12. How can you execute a Python script in IDLE? [U] / [CO2]
    Open IDLE.
    Write the Python program.
    Save the file with .py extension.
    Press F5 or select Run → Run Module to execute the script.

13. Define Unsupervised Learning and mention one real-life application. [A] / [CO3]
    - *Its a type of ML that works with unlabeled data to find hidden patterns or structures.*
    - *Real-life application: Customer segmentation in marketing*

14. What is the primary Python library used for importing and exporting data in Python? [K] / [CO2]

Pandas library.

15. Explain how Machine Learning is used in healthcare. [A] / [CO3]

- *Disease diagnosis using medical images.*
- *Predicting patient health risks.*
- *Personalized treatment recommendations.*
- *Drug discovery and development.*
- *Monitoring patient data and health records.*

16. What do you understand by 'Training Neural Networks'? [U] / [CO1]

- *Training Neural Networks means teaching the network using input data.*
- *The network adjusts weights and biases to reduce errors.*
- *It uses algorithms like backpropagation and gradient descent.*
- *The goal is to improve prediction accuracy.* ***(5X4=20)***

Part C

Essay Questions . Answer any 2 questions Each question carries 10 marks

17. Discuss how Import and Export data using python [U] / [CO2]

- *Importing data means reading data from files like CSV, Excel, or databases into Python.*
- *Exporting data means saving processed data from Python into files or databases.*
- *Python libraries used: Pandas, CSV, OpenPyXL.*
- *Common functions: read_csv(), read_excel(), to_csv(), to_excel().*
- *Helps in data analysis and machine learning tasks.*

18. How does model evaluation help in improving Supervised Learning techniques? [U] / [CO3]

- *Measures model accuracy and performance.*
- *Helps detect overfitting and underfitting.*
- *Improves prediction quality.*
- *Assists in selecting the best algorithm.*
- *Uses evaluation metrics like accuracy, precision, recall, and F1-score.*

19. What are the key obstacles encountered when deploying machine learning models? [An] / [CO1]

- *Data quality and availability issues.*
- *Model overfitting and underfitting.*
- *High computational cost.*
- *Security and privacy concerns.*
- *Difficulty in model integration with existing systems.*
- *Model maintenance and updates.*

20. Write a Python program to demonstrate the use of functions, loops, and conditional statements together. (any program) [A / [CO2]

eg:

```
def check_even(num):
if num % 2 == 0:
return "Even"
else:
return "Odd"
for i in range(1, 6):
result = check_even(i)
print(i, "is", result)
```

(10X2=20)